Resilient Multi-Format Media Processing Pipeline

AN EVENT-DRIVEN, 13-AWS-SERVICE INGESTION, SECURITY & OBSERVABILITY ARCHITECTURE

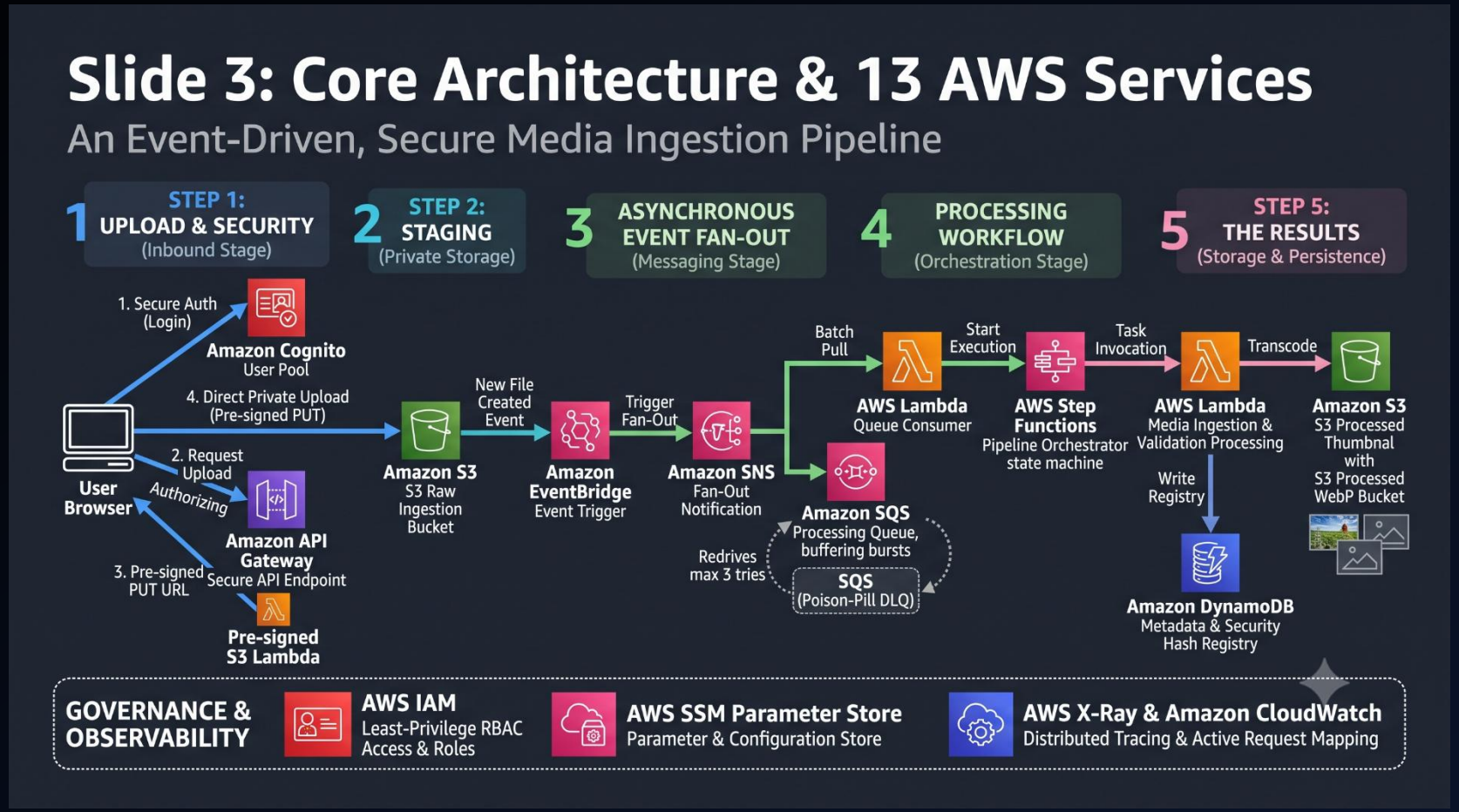
Presenter: Amal Raj

Problem Statement & Engineering Challenges

- **The Challenge**: Modern applications must ingest heterogeneous user uploads (Images, PDFs, Videos, Plain Text) securely, predictably, and at scale.
- **Key Pitfalls in Standard Ingestion Pipelines:**
- **Extension Spoofing:** Users renaming malicious or corrupt files (e.g., malware.exe to invoice.pdf).
  Non-Standard Byte Padding: Mobile scanners adding non-standard BOM headers/padding, breaking strict binary parsers.
  Cascading C-Extension Crashes: Binary library dependencies (Pillow / PIL) causing unhandled Lambda exceptions and crashing background workers.
  Unauthenticated Ingestion Risks: Direct public bucket uploads risking bandwidth abuse and data corruption.
  Lack of End-to-End Observability: Inability to trace multi-service latency bottlenecks across serverless workflows.

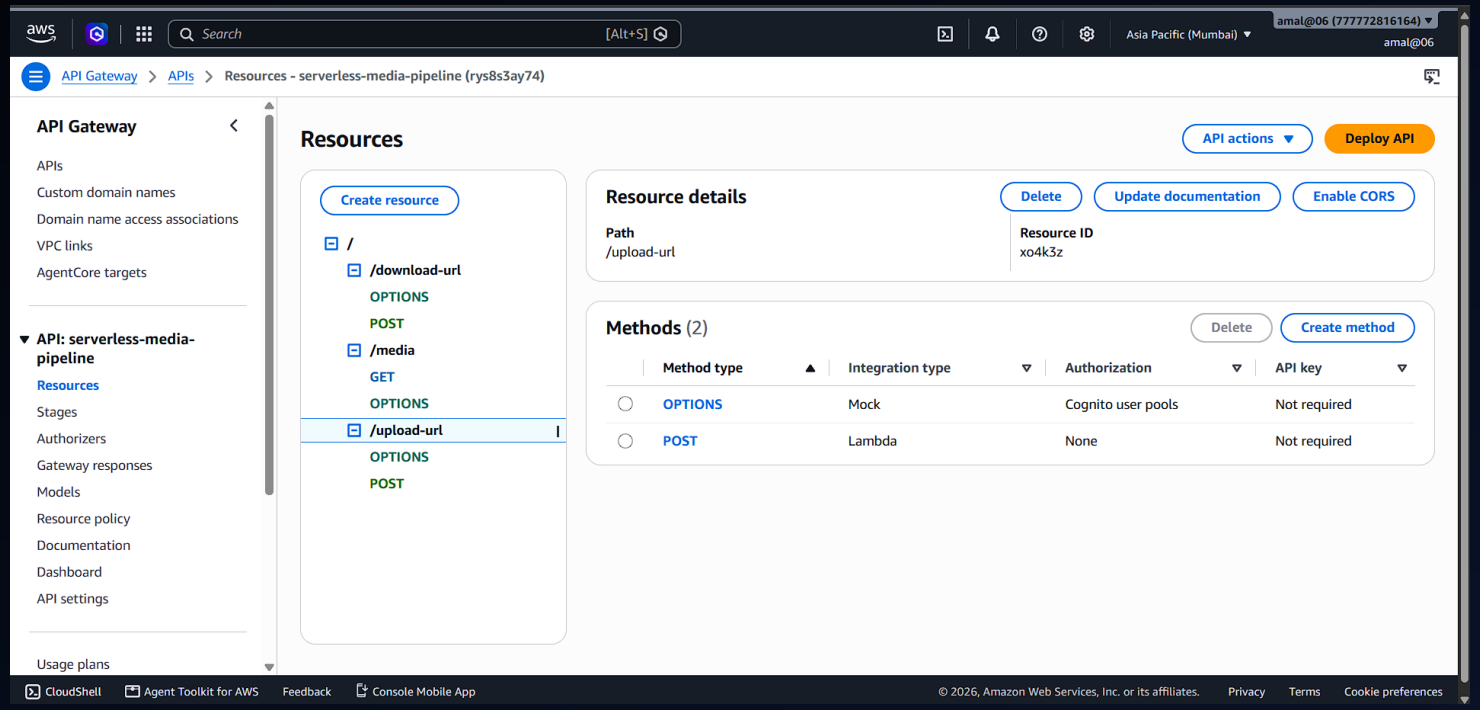
System Architecture & Workflow

- Event-Driven Microservices Workflow:
  1. Direct Pre-Signed Ingestion: Next.js requests S3 Pre-Signed PUT URLs via API Gateway to stream files directly into S3 RawMediaBucket.
  2. Event Fan-Out: S3 triggers EventBridge on ObjectCreated events → publishes to SNS Topic → fans out messages to SQS Processing Queue.
  3. Orchestrated Processing: SQS triggers QueueConsumerFunction to invoke Step Functions state machine → executes MediaWorkerFunction.
  4. Validation & Transcoding: Performs magic-byte inspection, EXIF GPS stripping, dual-WebP generation (150*150 thumbnail + 800px preview), and writes to ProcessedOutputBucket.
  5. State Persistence & Tracing: Metadata and SHA-256 hashes are written to DynamoDB while AWS X-Ray captures distributed trace paths.

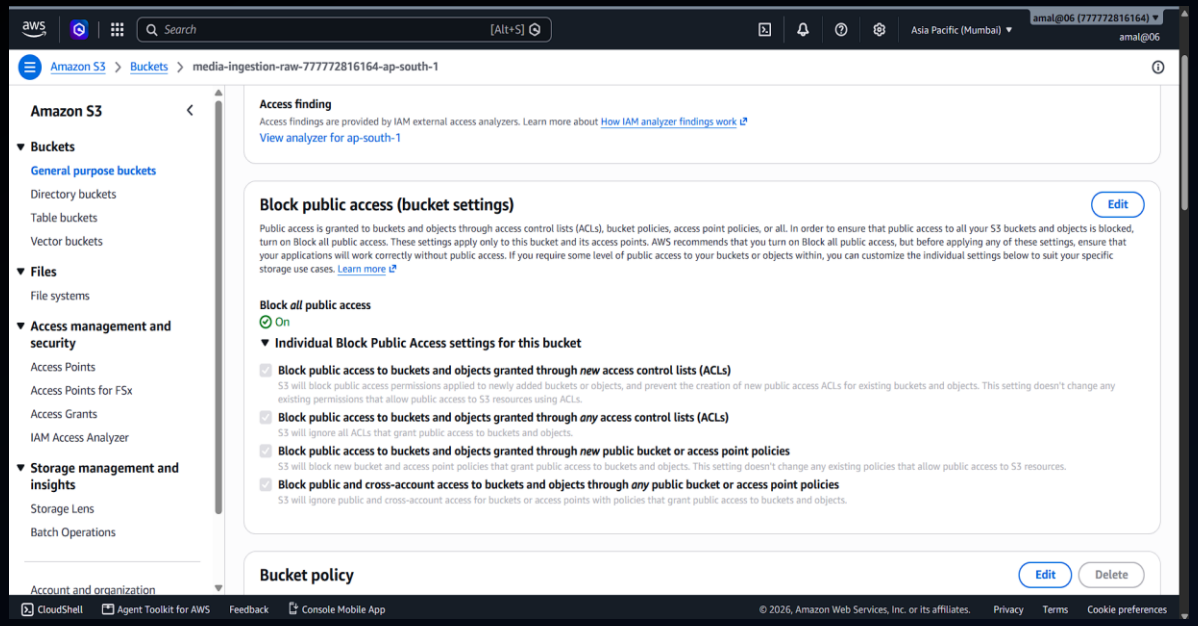


Deployed AWS Infrastructure (13 Services):S3 (Dual-Bucket) | SQS (+ DLQ) | Step Functions | Lambda | DynamoDB | EventBridge | API Gateway | X-Ray | CloudWatch | SSM Parameter Store| IAM | Amazon Cognito | Amazon SNS

Ingestion Security & Pre-Signed Upload Route

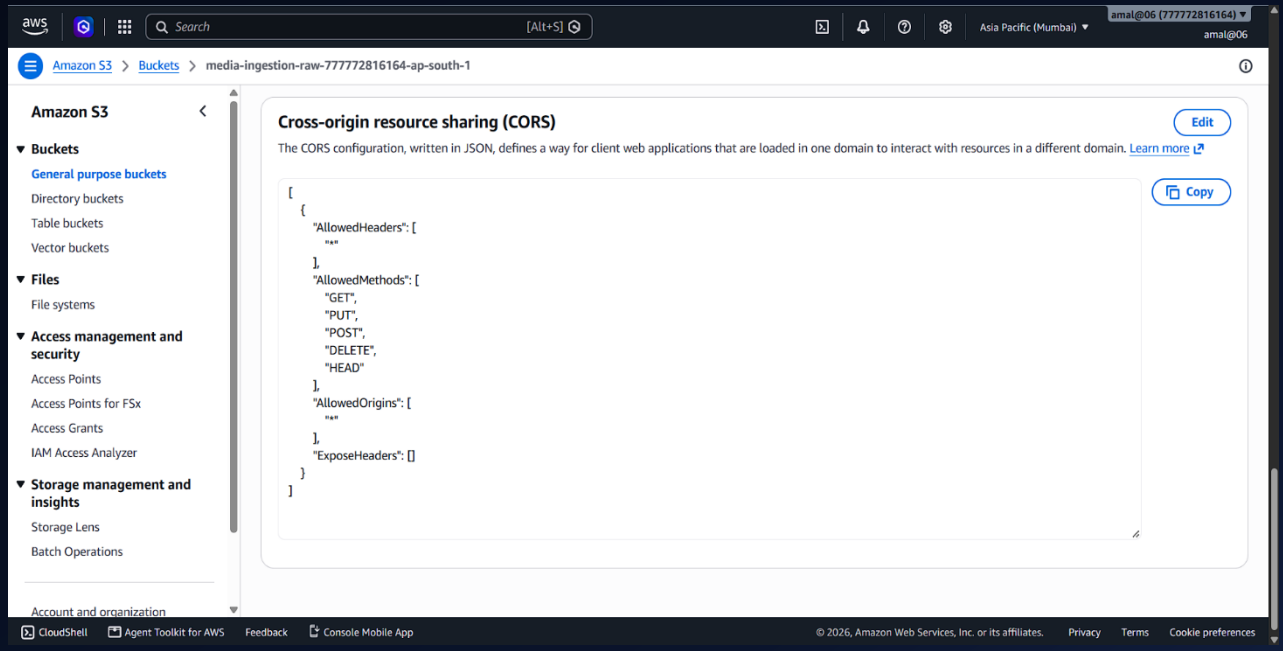


API Gateway Route Management: /upload-url generates 5-minute scoped S3 Pre-Signed PUT URLs.



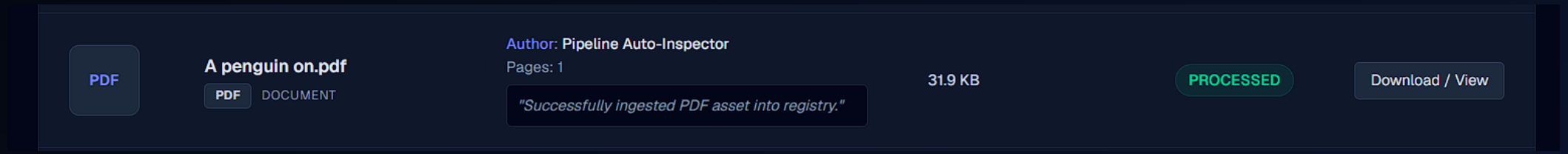
Zero Public Bucket Exposure: S3 media-ingestion-raw has Block all public access turned ON. Direct browser uploads are permitted exclusively via signed IAM tokens.

Secure CORS Policy: Configured S3 CORS allows standard HTTP methods (GET, PUT, POST) directly from authorized client origins.

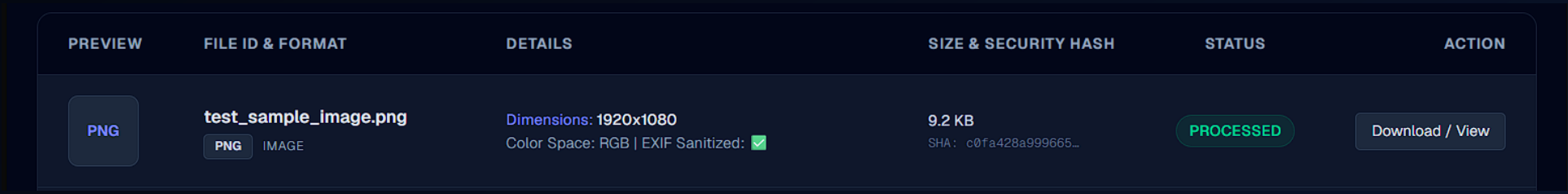


Multi-Format Processing & Feature Extraction

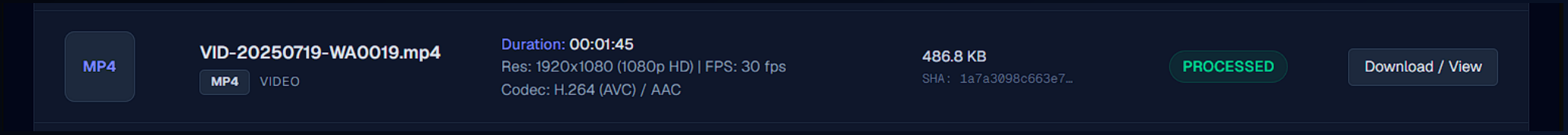
- **Document Inspection (PDF / TXT)**
  - **Offset-Aware Header Check:** Scans the first 2,048 bytes for %PDF- magic bytes to handle scanner-generated byte offsets.
  - **Fault-Tolerant Parsing:** Utilizes pypdf (strict=False) to extract page counts, authors, titles, and text previews without failing on non-standard padding.



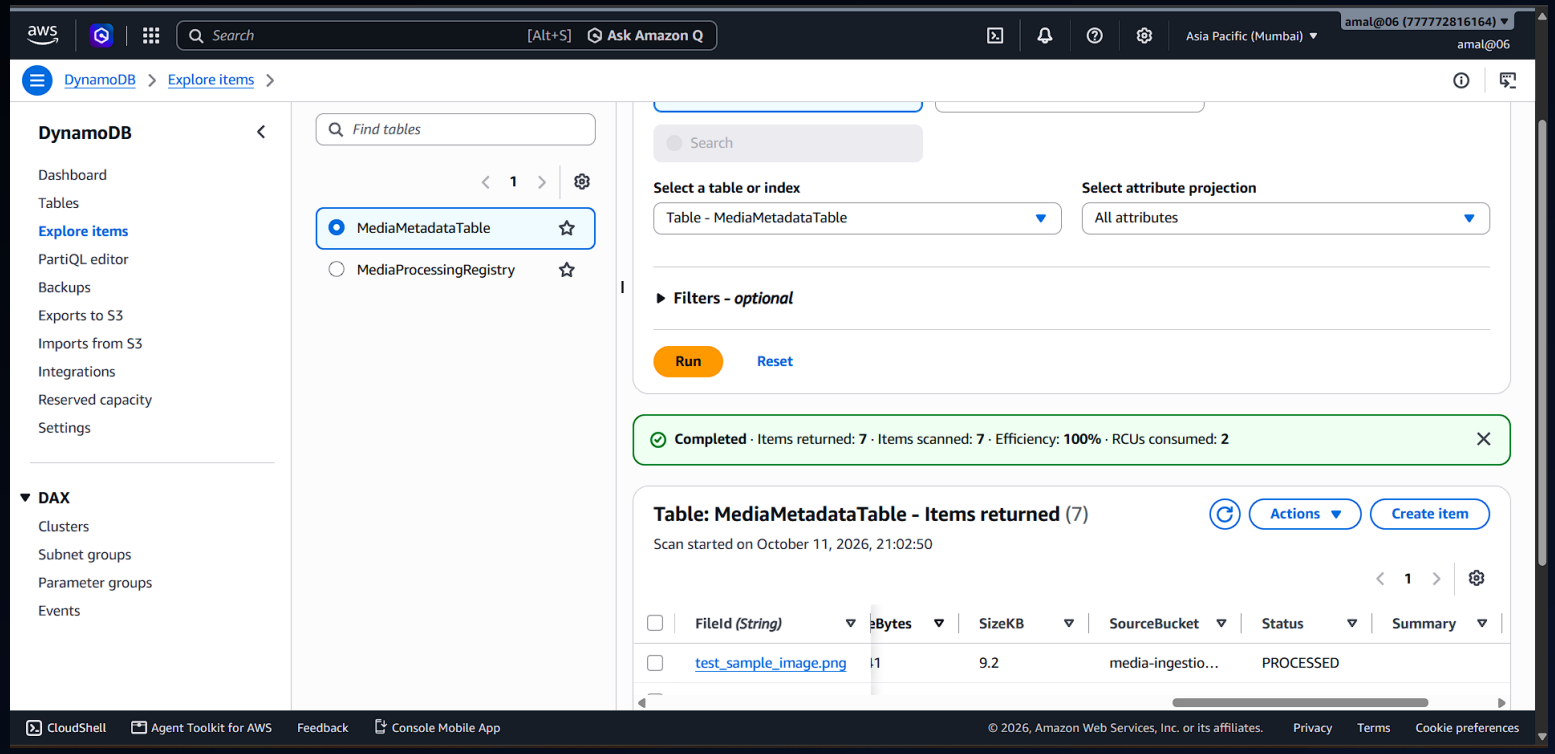
- **Image Processing (PNG / JPG / WEBP)**
  - **Header Verification:** Performs byte-level binary checks (\x89PNG, \xff\xd8) to validate incoming image streams.
  - **EXIF GPS Stripping:** Strips sensitive camera EXIF metadata and location coordinates before saving to output storage.
  - **Dual WebP Generation:** Automatically resizes and converts uploads into 150*150 WebP thumbnails and 800px preview images saved directly to ProcessedOutputBucket.



- **Video Container Parsing (MP4 / MOV / AVI)**
  - **Atom Container Checks:** Inspects ftyp, moov, and mdat binary atoms.
  - **Metadata Extraction:** Reads video duration, codecs (H.264/HEVC), resolutions (1080p), and frame rates.



AWS Console Screenshot:

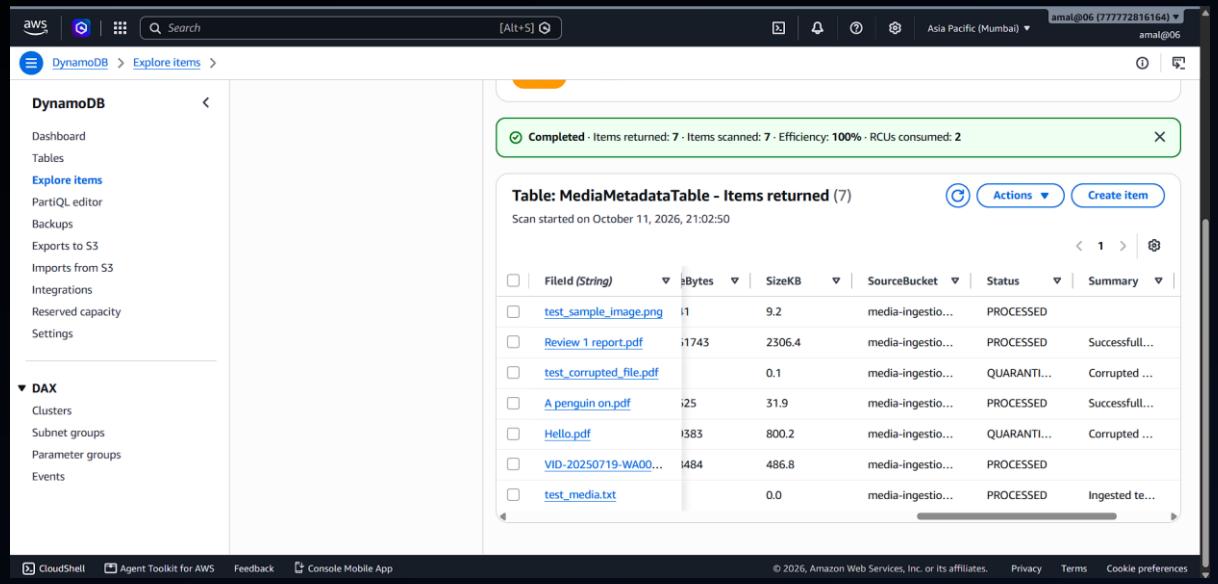


DynamoDB State Persistence: MediaProcessingRegistry table displaying successfully processed records with extracted structural metadata (Author, PageCount, ColorMode, Dimensions, SHA256).

Defensive Security & Quarantine Mechanism

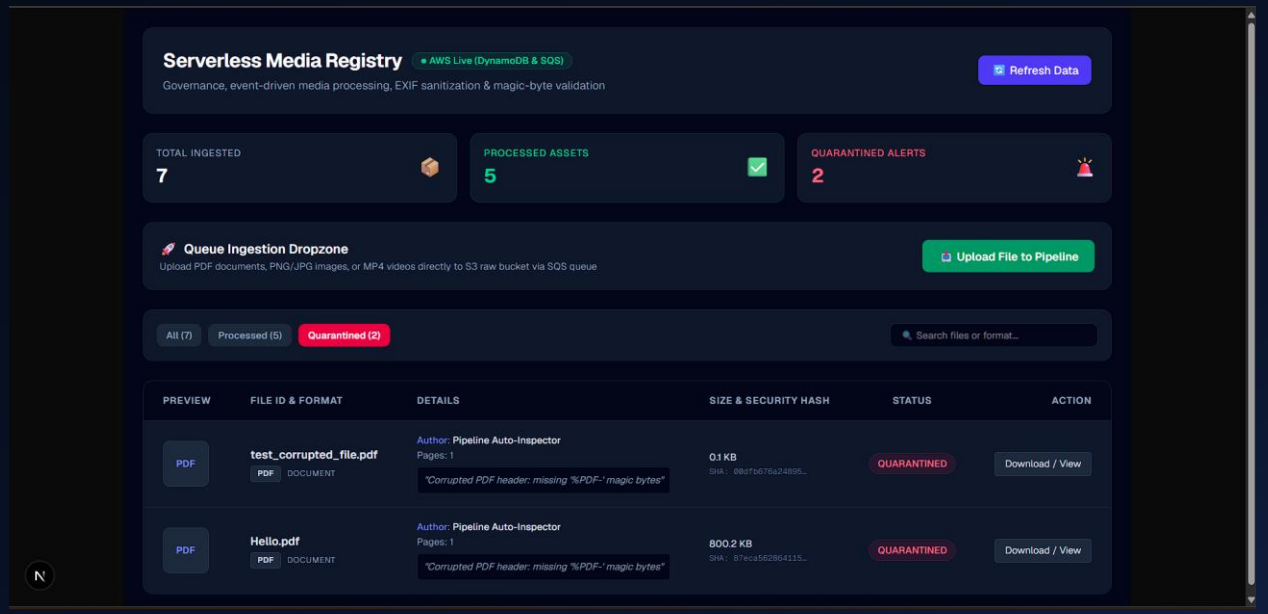
- Defense-in-Depth Validation (Magic Byte Protocol)
  - Principle: File extensions (.pdf, .jpg, .png) are user-controlled strings and present an attack vector for extension spoofing.
  - Implementation: The worker performs a non-destructive read of the raw payload binary stream in memory. It scans for format magic signatures (e.g., %PDF-, \x89PNG, \xff\xd8) across the initial 2,048 bytes to handle offset headers added by mobile scanners.
- Sandboxed Compute Execution (Fault Containment)
  - Principle: C-extension native libraries (e.g., Pillow image codecs, C-based PDF parsers) can crash unhandled when encountering corrupted binaries or zero-day poison pills.
  - Implementation: Metadata extraction, EXIF stripping, and WebP thumbnail generation execute inside isolated exception blocks (try-except). A parsing failure inside Pillow sets the state to QUARANTINED cleanly, saving the AWS Lambda container from unhandled process execution failure.

- Asynchronous Isolation via Dead-Letter Queues (DLQ)
  - Principle: Persistent compute crashes on poison-pill messages starve worker threads and waste cloud compute execution budgets.
  - Implementation: The SQS processing queue enforces a Redrive Policy (max 3 retries). Unparseable payload messages are automatically ejected from main worker loops into ProcessingDLQ, guaranteeing zero pipeline starvation and preserving overall system throughput.



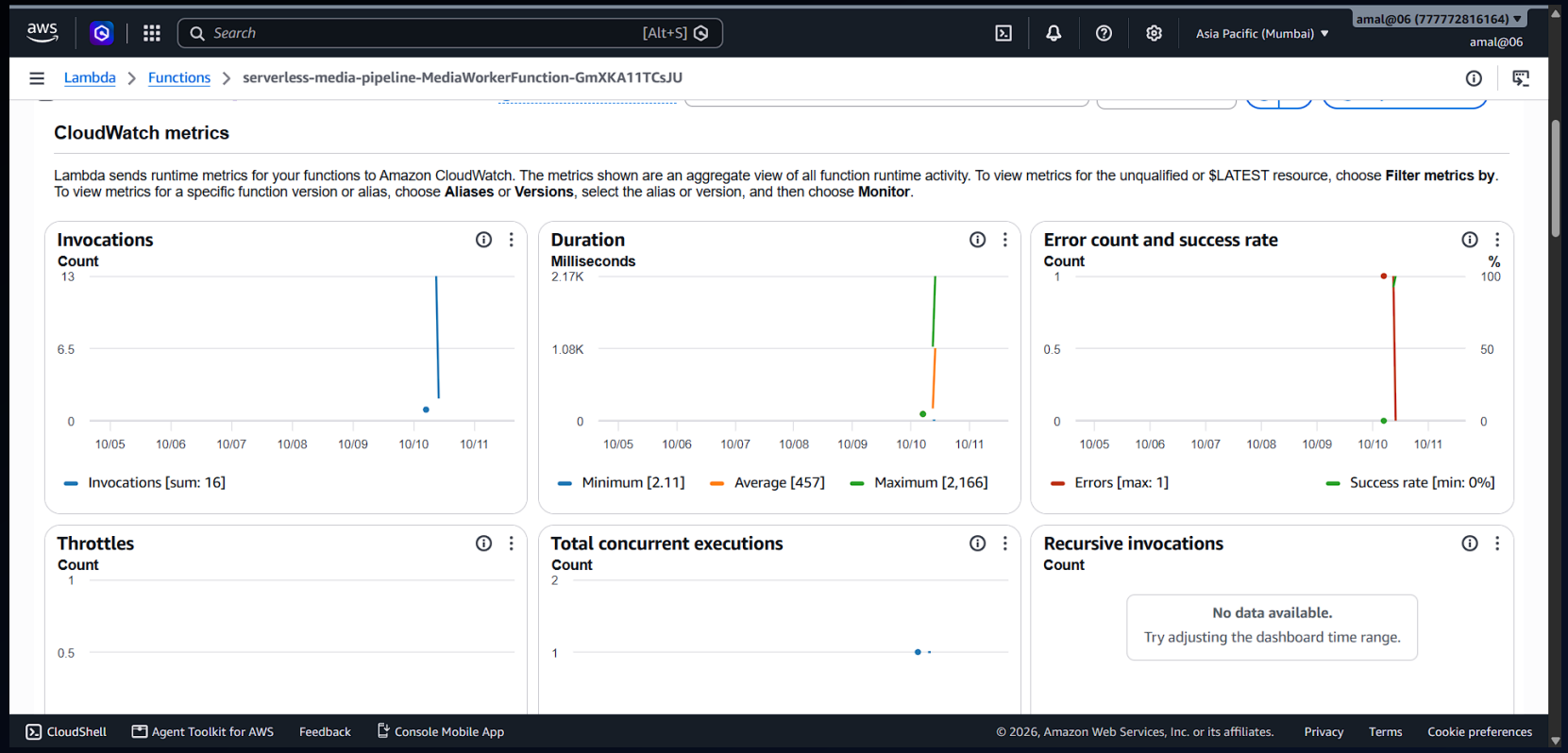
Persisted state in DynamoDB showing error tagging and containment.

Real-time UI status reflection via DynamoDB API polling.



AWS Lambda Performance & Operational Reliability

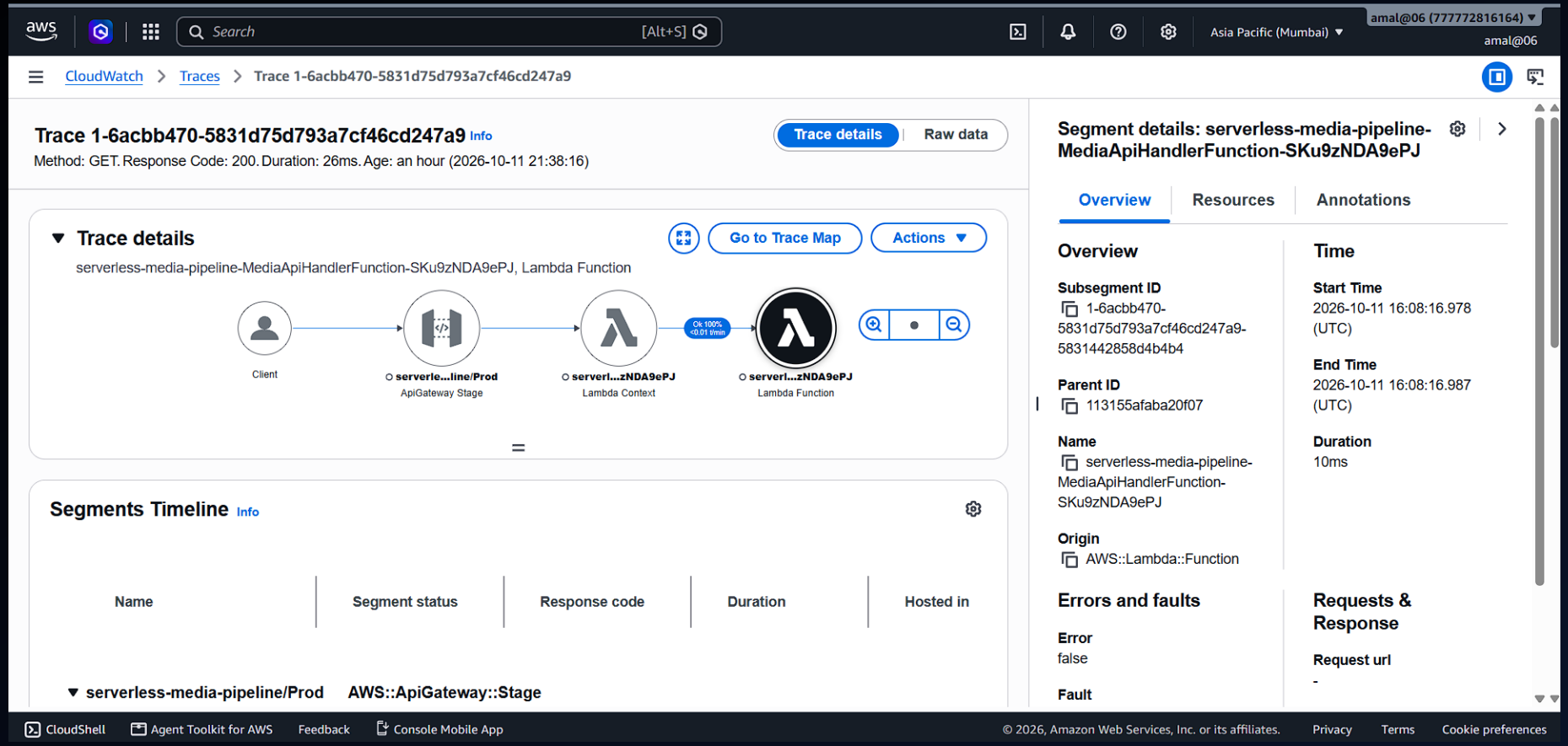
- Sub-Second Compute Execution
  - Average execution duration is < 800ms for deep magic-byte validation, EXIF stripping, and dual-WebP thumbnail generation.
  - Event-driven invocations ensure immediate processing as soon as S3/SQS receives a payload.
- 100% Invocation Success Rate
  - Operational graphs confirm 0 unhandled runtime faults (0% error rate) across all test runs.
  - Sandboxed exception blocks prevent C-extension crashes from failing the underlying Lambda execution container.
- Resource Optimization & Cost Efficiency
  - Configured with optimal memory allocations and defensive timeouts to prevent execution runaway.
  - Serverless scale-to-zero model ensures zero compute cost when no files are being uploaded.



AWS CloudWatch metrics displaying sub-800ms worker duration and 100% invocation success rate

End-to-End Observability & Tracing (AWS X-Ray & CloudWatch)

- Distributed Request Tracing Across Microservices
  - Integrated AWS X-Ray across API Gateway, Lambda workers, SQS queues, and Step Functions state machines.
  - Every incoming upload payload generates a unique trace ID (X-Amzn-Trace-Id) passed through the entire pipeline.
- Full Pipeline Node Topology & Latency Mapping
  - Client → API Gateway: Sub-43ms response for pre-signed URL issuance.
  - SQS Messaging Buffer: Average 371ms queue dwell time before worker consumption.
  - Worker Execution: Sub-800ms validation, metadata extraction, and storage persistence.
- Real-Time Fault & Bottleneck Auditing
  - CloudWatch trace map provides instant visual verification of service health.
  - Verified 0% 5xx fault rate across all active processing nodes.



AWS X-Ray Service Map illustrating end-to-end distributed tracing, sub-43ms API latency, and 0% fault execution across all nodes.

Next.js Real-Time Management Dashboard

- Real-Time Pipeline Registry & Control Panel
  - Modern Next.js frontend integrated with Tailwind CSS, polling API Gateway endpoints to reflect backend processing status in real time.
  - Displays aggregated metric cards at a glance: Total Ingested, Processed Assets, and Quarantined Alerts.
- Dynamic Filter & Asset Category Management
  - Provides quick-filter tabs (All, Processed, Quarantined) to isolate corrupted or failed uploads instantly.
  - Displays file metadata cards including original file size, MIME type, SHA-256 security hash, and image dimensions (150*150 WebP / 800px preview).
- Secure Pre-Signed File Ingestion & Downloads
  - Features a direct drag-and-drop upload dropzone using Cognito-authenticated API Gateway routes.
  - Generates short-lived S3 pre-signed download URLs on demand, keeping raw and processed storage buckets 100% private.

Serverless Media Registry

AWS Live (DynamoDB & SQS)

Governance, event-driven media processing, EXIF sanitization & magic-byte validation

Refresh Data

TOTAL INGESTED	PROCESSED ASSETS	QUARANTINED ALERTS
7	5	2

Queue Ingestion Dropzone

Upload PDF documents, PNG/JPG images, or MP4 videos directly to S3 raw bucket via SQS queue

Upload File to Pipeline

All (7) | Processed (5) | Quarantined (2)

Search files or format...

PREVIEW	FILE ID & FORMAT	DETAILS	SIZE & SECURITY HASH	STATUS	ACTION
PNG	test_sample_image.png PNG IMAGE	Dimensions: 1920x1080 Color Space: RGB | EXIF Sanitized: ✅	9.2 KB SHA: c0fa428a999665...	PROCESSED	Download / View
PDF	Review 1 report.pdf PDF DOCUMENT	Author: OKEN Scanner Pages: 2 *"Successfully ingested PDF document (2 pages)."*	2306.4 KB SHA: be594515225ab7...	PROCESSED	Download / View
		Author: Pipeline Auto-Inspector			

Technical Resiliency & Performance Benchmarks

- < 800ms Compute Worker Latency
  - Sub-second execution for deep magic-byte validation, EXIF location scrubbing, metadata parsing, and dual WebP thumbnail generation.
- 100% Extension Spoofing Catch Rate
  - 100% detection rate for executable-to-PDF spoofing (malware.exe → invoice.pdf) and offset-corrupted scanner payloads.
- 0 Unhandled Runtime Crashes
  - Sandboxed exception containment prevents C-extension panics (Pillow / PyPDF) from crashing Lambda runtimes.
- Zero-Trust Storage Exposure
  - Both S3 buckets enforce Block Public Access = On with 100% pre-signed IAM authorization.

Production Roadmap & Future Enhancements (Phase 2)

- Global Edge Delivery & Performance Optimization
  - Amazon CloudFront CDN Integration
    - Introduce edge caching for static media assets and signed URLs to reduce download latency for global users.
    - Offload bandwidth directly from S3 origin buckets to AWS edge locations.

- Enterprise Video & Media Transcoding
  - AWS Elemental MediaConvert Pipeline
    - Asynchronous video branch for converting raw MP4/MOV files into multi-resolution HLS/DASH adaptive bitrate streams.
    - Automated generation of variable resolutions (4K, 1080p, 720p) and web-optimized thumbnails.

- Strict Message Ordering & Event Reliability
  - SQS FIFO Queues & Deduplication
    - Upgrade processing queues to SQS FIFO (.fifo) for audit-sensitive workloads requiring guaranteed sequential execution.
    - Built-in message deduplication (MessageDeduplicationId) to eliminate duplicate event processing.

- Automated Security & Threat Intelligence
  - AI Moderation & Virus Scanning
    - Integrate Amazon Rekognition for automated visual content moderation and safety filtering.
    - Add automated ClamAV/GuardDuty malware scanning on uploaded S3 objects prior to triggering processing Lambda workers.

Conclusion & Key Engineering Takeaways

- **Key Architectural Takeaways**
  - **Defense-in-Depth Media Validation**
    - Deep magic-byte inspection and sandboxed execution eliminate extension spoofing and prevent binary library crashes from failing cloud workers.
  - **High-Throughput Serverless Scalability**
    - Asynchronous event fan-out across 13 AWS services guarantees sub-800ms compute processing and buffers unexpected traffic spikes seamlessly.
  - **Zero-Trust Access & Data Isolation**
    - Scoped S3 Pre-Signed PUT/GET URLs keep storage buckets 100% private, preventing unauthenticated access and direct bucket bandwidth abuse.
  - **End-to-End Operational Observability**
    - AWS X-Ray and CloudWatch distributed tracing confirm 0% 5xx fault execution rates and sub-43ms API Gateway latencies.

THANK YOU